UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

SEDE SANTO DOMINGO DE LOS TSACHILAS
CARRERA DE ITIN
Periodo Marzo-Agosto 2026

4817

Examen3

Asignatura: **MÉTODOS NUMÉRICOS** **NRC: 29528**
Docente: RODRIGO BASTIDAS Semestre: 3 ro Paralelo: "B"
Nombre del estudiante: Jeffry Vera
C.c.: 2350511115 **Fecha:** 30/07/2026

Indicaciones

- **Tiempo de duración de la prueba 1.5 horas**
- **Desarrollo de los ejercicios a lápiz o bolígrafo con número y letras legibles y claras**
- **Cualquier acto de copia será sancionado.**

1. Resolver la siguiente ecuación diferencial por diferencias finitas **(4 p)**:

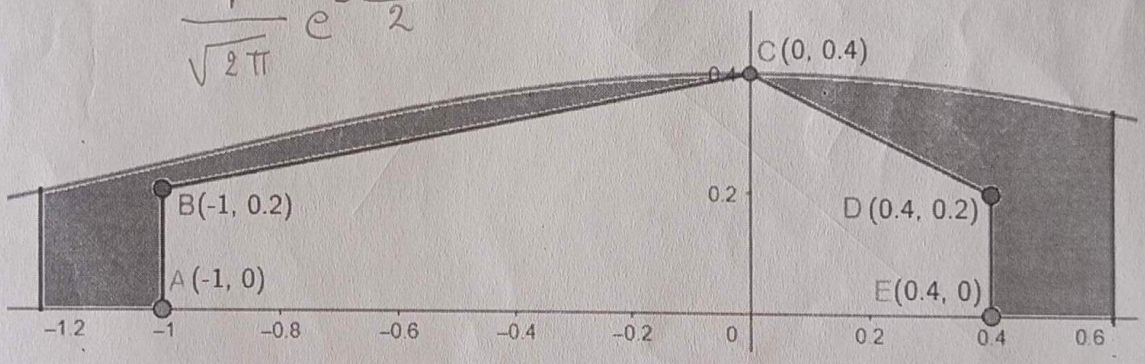
$$x^2y'' - xy' - x^3y = \text{sen}(x^2)$$

$$y(3) = 1, y(5) = 1, \qquad 3 \leq x \leq 5$$

Asuma n =5 y plantee el sistema de ecuaciones.

2. Dada la siguiente figura, encontrar el área entre la curva y el polígono. Utilice el método Simpson 3/8 para la curva (función distribución normal) con n=6 y para el polígono el método del trapecio **(3 p)**

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$



Nota: Presente su respuesta final con **cuatro decimales, aproximación por redondeo**

$$\int \approx \frac{h}{3}\left(y_0 + 3(y_1 + y_2)^2(y_3 + y_4)^2(y_5 + y_6)\right) + y_n$$

$$y' = \frac{-y_{i-1} + y_{i+1}}{2h} \qquad y'' = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2}$$

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① $x^2y'' - xy' - x^3y = \operatorname{sen}(x^2)$

$$y'' = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2}$$

$$y' = \frac{-y_{i-1} + y_{i+1}}{2h}$$

$$x^2\left(\frac{y_{i-1} + 2y_i + y_{i+1}}{h^2}\right) - x\left(\frac{-y_{i-1} + y_{i+1}}{2h}\right) - x^3y = \operatorname{sen}(x^2) \quad * 2h^2 \qquad 3 \le x \le 5$$

$$2x^2(y_{i-1} - 2y_i + y_{i+1}) - xh(-y_{i-1} + y_{i+1}) - 2h^2x^3y_i = 2h^2\operatorname{sen}(x^2)$$

$$y_{i-1}(2x^2 + xh) + y_i(-4x^2 - 2h^2x^2) + y_{i+1}(2x^2 - xh) = 2h^2\operatorname{sen}(x^2)$$

Para $i = 1$ $\quad x_1 = 3 + 0.4 = 3.4$ $\qquad h = \frac{5-3}{5} = 0.4$

$y_0 = 1 = (2(3.4)^2 + (3.4)(0.4)) = 24.4800$

$y_1 = -4(3.4)^2 - 2(0.4)^2(3.4)^2 = -49.9392$

$y_2 = 2(3.4)^2 - (3.4)(0.4) = 21.76$

$2(0.4)^2\operatorname{Sen}(3.4)^2 = 0.02089$

-24.46711 $\quad -58,81x$ $\quad -24,75$

Ecuación 1 $= -49.9392y_1 + 21.76y_2 = -24.46711$

Para $i = 2$ $\quad x_2 = 3 + 2(0.4) = 3.8$

$y_1 = 2(3.8)^2 + (3.8)(0.4) = 30.4000$

$y_2 = -4(3.8)^2 - 2(0.4)^2(3.8)^2 = -62.3808$

$y_3 = 2(3.8)^2 - (3.8)(0.4) = 27.36$

$2(0.4)^2\operatorname{Sen}(3.8)^2 = 0.119798$

$-75,31$ $\quad 0,3054$

Ecuacion 2 $= 30.4000y_1 - 62.3808y_2 + 27,36y_3 = 0.119798$

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Para $i=3$ $\quad x_3 = x_0 + 3h = 4.2$

$y_2 = 2(4.2)^2 + (4.2)(0.4) = 36.9600$

$y_3 = -4(4.2)^2 - 2(0.4)^2(4.2)^2 = -76.2048$

$y_4 = 2(4.2)^2 - (4.2)(0.4) = 33.6000$

$2(0.4)^2 \operatorname{Sen}(3.8)^2 = 0.243086$

-94.26 -0,2393

Ecuación 3 $= 36.9600 y_2 - 76.2048 y_3 + 33.6000 y_4 = 0.243086$

Para $i=4$ $\quad x_4 = x_0 + 4h = 4.6$

$y_3 = 2(4.6)^2 + (4.2)(0.4) = 44.16$

$y_4 = -4(4.6)^2 - 2(0.4)^2(4.6)^2 = -91.4112$

$y_5 = 2(4.6)^2 - (4.6)(0.4) = 40.4800$

$2(0.4)^2 \operatorname{Sen}(4.6)^2 = 0.789937$

115.78 -40,24

Ecuación 4 $= 44.16 y_3 - 91.4112 y_4 + 40.4800 y_5 = 0.789937$

Ec1 $\quad -49.9392 y_1 + 21.7600 y_2 = 24.46711$

Ec2 $\quad 30.4000 y_1 - 82.3808 y_2 + 27.3600 y_3 = 0.119748$

Ec3 $\quad 36.9600 y_2 - 76.2048 y_3 + 33.600 y_4 = 0.243086$

Ec4 $\quad 44.16 y_3 - 91.4112 y_4 + 40.4800 y_5 = 0.789937$

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Trapecio

$$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Area 1 $[-1, 0)$ $\quad x_0 \quad x_1$

$h = 0 + 1 = 1$

$x_1 - x_0$

$0 +$

$$f(x_0) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(-1)^2}{2}} = 0.2419$$

$$f(x_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{0^2}{2}} = 0.3989$$

$$\text{Area } 1 = 1\left(\frac{0.2419 + 0.3989}{2}\right) = 0.3204$$

Area 2 $[0, 0.4)$ $\quad x_0 \quad x_1$

$h = 0.4 - 0 = 0.4$

$$f(x_0) = \frac{1}{\sqrt{2\pi}} e^{-\frac{0^2}{2}} = 0.3989$$

$$f(x_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(0.4)^2}{2}} = 0.3682$$

0.6408

0.7671

$$\text{Area } 2 = 0.4\left(\frac{0.3989 + 0.3682}{2}\right) = 0.38355$$

Area 1 + Area 2 = 0.3204 + 0.38355 = 0.70395

Total = 0.70395 //

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Simpson $[-1.2, 0.6]$ $n = 6$

$$h = \frac{0.6 + 1.2}{6} = 0.3$$

$$\int_{-1.2}^{0.6} \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} = \frac{3h}{8}\left(y_0 + 3(y_1 + y_4) + 3(y_2 + y_5) + 2(y_3) + y_6\right)$$

$x_0 = -1.2$ $\quad y_0 = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_0^2}{2}} = 0.1941$

$x_1 = x_0 + h = -0.9$ $\quad y_1 = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_1^2}{2}} = 0.2660$

$x_2 = x_0 + 2h = -0.6$ $\quad y_2 = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_2^2}{2}} = 0.3332$

$x_3 = x_0 + 3h = -0.3$ $\quad y_3 = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_3)^2}{2}} = 0.3813$

$x_4 = x_0 + 4h = 0$ $\quad y_4 = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_4)^2}{2}} = 0.3989$

$x_5 = x_0 + 5h = 0.3$ $\quad y_5 = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_5)^2}{2}} = 0.3813$

$x_6 = 0.6$ $\quad y_6 = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_6)^2}{2}} = 0.3332$

$$\int_{-1.2}^{0.6} \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \approx \frac{3(0.3)}{8}\left(0.1941 + 3\left((0.2660) + (0.3989)\right) + 3\left((0.3332) + (0.3813)\right) + 2(0.3813) + 0.3332\right)$$

1.9947 $\quad$ 2.1435 $\quad$ 0.7626

$$\approx \frac{3(0.3)}{8}(5.4281)$$

$$\int_{-1.2}^{0.6} \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} \approx 0.610661$$

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Area Trapecio = 0.70395

Area Sim $\frac{3}{8}$ = 0.610661

0,42

An - A

1,8

Area Total = 0.70395 + 0.610661 = 1.3146

>1

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